

An exposition of a computer-assisted proof of the bound $\Lambda \leq 0.1787854$ for the de Bruijn–Newman constant

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Abstract

We give a self-contained mathematical exposition of a claimed computer-assisted proof, due to J. Gomila, of the bound

$$\Lambda \leq \frac{893927}{5000000} = 0.1787854$$

for the de Bruijn–Newman constant Λ , improving the bound $\Lambda \leq 0.22$ of the Polymath 15 project and the current record $\Lambda \leq 0.2$ of Platt–Trudgian. The argument is intended to be unconditional: it instantiates the criterion of Polymath 15 [Theorem 1.2, arXiv:1904.12438v2] at the exact parameters $X = 6000000185827$, $t_0 = 129/800$, $y_0^2 = 87677/2500000$, supplying its three hypotheses by, respectively, the Platt–Trudgian finite verification of the Riemann hypothesis, a hybrid finite-scan/analytic-tail nonvanishing theorem for H_{t_0} , and an interval-arithmetic barrier certificate. The proof has not been peer reviewed, and the correctness of neither the analytic reductions nor the extensive interval computations has been independently confirmed. All statements quoted here from the published literature have been checked against the arXiv sources of the cited papers. The steps of the argument that remain unverified — its own analytic lemmas and its machine computations — are marked throughout as numbered *gap* / *review points*, and are collected in Section 8.

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*Referee’s exposition of an unpublished proof by J. Gomila (private repository `dbn-lambda-01787854-candidate-audit`, “sealed referee snapshot,” 2026). This document describes the argument for the purpose of independent review; it does not currently certify its correctness. All gaps and steps that this expositor has not verified are marked explicitly.

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1 Introduction

1.1 The de Bruijn–Newman constant

Let Φ denote the super-exponentially decaying function

$$\Phi(u) = \sum_{n=1}^{\infty} (2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u}) \exp(-\pi n^2 e^{4u}),$$

and for $t \in \mathbb{R}$ define the entire functions

$$H_t(z) = \int_0^{\infty} e^{tu^2} \Phi(u) \cos(zu) du. \tag{1}$$

The function H_0 is a normalization of the Riemann ξ -function: with $\xi(s) = \frac{s(s-1)}{2} \pi^{-s/2} \Gamma(s/2) \zeta(s)$ one has $H_0(x) = \frac{1}{8} \xi(\frac{1}{2} + \frac{ix}{2})$, so that the Riemann hypothesis (RH) is the statement that all zeros of H_0 are real. De Bruijn [1] showed that if H_t has only real zeros for some t , the same holds for all larger t ; Newman [4] showed that for some t the function H_t has a non-real zero. Hence there is a constant $\Lambda \in \mathbb{R}$, the *de Bruijn–Newman constant*, such that H_t has only real zeros if and only if $t \geq \Lambda$. RH is equivalent to $\Lambda \leq 0$.

Newman conjectured $\Lambda \geq 0$, which was proved by Rodgers and Tao [7]. On the other side, a long sequence of numerical upper bounds (de Bruijn’s $\Lambda \leq 1/2$, then [2, 3] and others) led to the bound

$$\Lambda \leq 0.22 \tag{2}$$

of the Polymath 15 project [6]. Subsequently Platt and Trudgian observed [5, Corollary 2] that their verification of RH to height $3 \cdot 10^{12}$ (Section 3 below), combined with the second row of the parameter table in [6, Table 1, p. 65] (which requires verified height $H > 2.51 \cdot 10^{12}$), yields the current record

$$\Lambda \leq 0.2. \tag{3}$$

Thus at present $0 \leq \Lambda \leq 0.2$. Platt and Trudgian further note that the next entry of the same table would give $\Lambda < 0.19$, but only conditionally on a verified height slightly above 10^{13} , beyond their theorem; the bound (5) below therefore goes past both the published record (3) and that conditional entry, while consuming only the established $3 \cdot 10^{12}$ verification height.

1.2 Main result

This paper describes a claimed computer-assisted proof, due to J. Gomila, of the following improvement of (3).

Theorem 1.1 (Theorem; **unreviewed**).

$$\Lambda \leq \frac{893927}{5000000} = 0.1787854.$$

The proposed proof instantiates the barrier criterion of [6, Theorem 1.2] at the exact parameters

$$X = 6000000185827, \quad t_0 = \frac{129}{800} = 0.16125, \quad y_0^2 = \frac{87677}{2500000} = 0.0350708, \tag{4}$$

with $y_0 > 0$, so that

$$t_0 + \frac{y_0^2}{2} = \frac{893927}{5000000} = 0.1787854. \tag{5}$$

One checks the elementary inequalities $0 < t_0 < \frac{1}{2}$, $0 < y_0^2 < 1 - 2t_0 = \frac{271}{400}$, and $y_0^2 + 2t_0 = \frac{893927}{2500000} < 1$, which place the parameters in the domain of the criterion.

1.3 Status, and how gaps are marked

The claimed proof combines:

1. two published theorem inputs — the Polymath 15 criterion and effective approximation [6], and the Platt–Trudgian finite verification of RH [5];
2. a collection of new lemmas (reproduced with proofs below), which bridge the published estimates to the quantities computed by the software; and
3. a large body of rigorous-interval-arithmetic certificates (FLINT/Arb in C, plus corroborating Python interval implementations), totalling 3,149,013 certified finite “window” rows, a standalone tail computation, and an 883-prism barrier certificate.

The statements quoted from the published inputs in item (1) — Theorems 1.2 and 1.3 of [6] together with the displays (14), (20)–(24), (71)–(72), Proposition 6.6 and Lemma 8.4 of that paper, and Theorem 1 of [5] — have been checked verbatim against the arXiv source files of the cited papers (1904.12438v2 and 2004.09765v1, to which all equation and theorem numbers below refer). The one point where the current proof’s usage deliberately departs from a cited display, namely equation (24) of [6], is discussed in Remark 2.2. Items (2)–(3) have not, with one exception, been independently verified: throughout the paper, every such unverified step is flagged by a framed, numbered *gap / review point*. (The exception is the main finite computation, Proposition 4.3, which has been reproduced on an independent toolchain with its program reviewed; its proof is given in Section 4.3.) These fall into two categories: **(A)** reliance on a machine computation, and **(B)** an analytic argument that requires independent checking. The full list is collected in Section 8.

2 The Polymath criterion and effective approximation

2.1 The barrier criterion

The logical skeleton of the proof is the following criterion.

Theorem 2.1 (Polymath [6, Theorem 1.2]). *Let $X > 0$, $0 < t_0 < \frac{1}{2}$ and $0 < y_0^2 < 1 - 2t_0$. Suppose that:*

(i) **(Verified height.)** *There are no zeros $\zeta(\sigma + iT) = 0$ with*

$$\frac{1 + y_0}{2} \leq \sigma \leq 1, \quad 0 \leq T \leq \frac{X}{2}.$$

(ii) **(Final-time right region.)** *There are no zeros $H_{t_0}(x + iy) = 0$ with*

$$x \geq X + \sqrt{1 - y_0^2}, \quad y_0 \leq y \leq \sqrt{1 - 2t_0}.$$

(iii) **(Intermediate-time barrier.)** *There are no zeros $H_t(x + iy) = 0$ with*

$$X \leq x \leq X + \sqrt{1 - y_0^2}, \quad \sqrt{y_0^2 + 2(t_0 - t)} \leq y \leq \sqrt{1 - 2t}, \quad 0 \leq t \leq t_0.$$

Then

$$\Lambda \leq t_0 + \frac{y_0^2}{2}.$$

Sections 3, 4–5, and 6 below describe the proof’s suppliers for hypotheses (i), (ii), and (iii) respectively, at the parameters (4).

For hypothesis (i) one uses the normalization connecting H_0 to ζ : a zero $H_0(x + iy) = 0$ gives $\xi(\frac{1-y+ix}{2}) = 0$; the functional equation $\xi(s) = \xi(1-s)$ followed by complex conjugation produces the representative zero $\xi(\frac{1+y+ix}{2}) = 0$ with $\sigma = \frac{1+y}{2} \geq \frac{1+y_0}{2}$ and ordinate $T = \frac{x}{2}$, which is the region excluded in (i). (The proof verifies the affine algebra of this “sign map” by an exact-rational script; the functional equation and the reality symmetry $\overline{\xi(\bar{s})} = \xi(s)$ are the classical inputs.)

2.2 The effective approximation

For $t > 0$ write, following [6],

$$g_t(z) = \frac{H_t(z)}{B_t(z)}, \quad (6)$$

where B_t is an explicit normalizing factor: in the notation of [6], $B_t(x + iy) = M_t\left(\frac{1+y-ix}{2}\right)$, where $M_t(s) = \exp\left(\frac{t}{4}\alpha(s)^2\right)M_0(s)$ and M_0 is defined in [6, (6)] as the exponential of an explicit holomorphic branch of its logarithm on $\mathbb{C} \setminus (-\infty, 1]$ (see [6, (7)]). In particular M_0 , and hence M_t , takes values in $\mathbb{C} \setminus \{0\}$ on its domain; since the argument $\frac{1+y-ix}{2}$ is non-real whenever $x \neq 0$, it avoids the branch cut, and therefore $B_t(x + iy) \neq 0$ for all $t \geq 0$, $x > 0$ and $y > 0$, as noted explicitly in [6, §1, after (11)]. This nonvanishing is used below both in the reduction of zero-freeness of H_t to lower bounds on $|f_t|$ and in the argument-principle step of the barrier certificate (Section 6). Theorem 1.3 of [6] provides, in the region

$$x \geq 200, \quad 0 \leq y \leq 1, \quad 0 < t \leq \frac{1}{2}, \quad (7)$$

an approximation

$$g_t(x + iy) = f_t(x + iy) + O_{\leq}(e_A + e_B + e_{C,0}), \quad (8)$$

where $O_{\leq}(E)$ denotes a quantity of absolute value at most E , the main term is the finite sum (equation (14) of [6])

$$f_t(x + iy) = \sum_{n=1}^N b_t(n) n^{-s_*} + \gamma \sum_{n=1}^N n^y b_t(n) n^{-\overline{s_*} - \kappa}, \quad b_t(n) = \exp\left(\frac{t}{4} \log^2 n\right), \quad (9)$$

(we write $b_t(n)$ for the quantity denoted b_n^t in [6]; all other symbols in (9) agree with the notation there, including the placement of κ alongside the conjugated exponent $\overline{s_*}$ in the second sum) with:

- the window index $N = \lfloor \sqrt{\frac{x}{4\pi} + \frac{t}{16}} \rfloor$;
- explicitly defined quantities $\gamma = \gamma(x, y, t)$, $s_* = s_*(x, y, t)$, $\kappa = \kappa(x, y, t)$ satisfying (by equations (20)–(22) of [6]) the bounds

$$|\gamma| \leq e^{0.02y} \left(\frac{x}{4\pi}\right)^{-y/2}, \quad \operatorname{Re} s_* \geq \frac{1+y}{2} + \frac{t}{4} \log \frac{x}{4\pi} - \frac{t}{2x^2} \left(1 - 3y + \frac{4y(1+y)}{x^2}\right)_+, \quad |\kappa| \leq \frac{ty}{2(x-6)}; \quad (10)$$

- and error terms $e_A, e_B, e_{C,0}$ with explicit majorants (equations (71)–(72) and (23)–(24) of [6]).

Since $B_t \neq 0$, nonvanishing of H_t on a region follows from the strict inequality $|f_t| > e_A + e_B + e_{C,0}$ there; this implication is itself stated in [6] as Corollary 1.4 (“Criterion for non-vanishing”).

2.3 The conservative error constant 10.50

The current proof deliberately does *not* use the constant 10.44 displayed in equation (24) of [6] for the majorization of $e_{C,0}$. Instead it derives a slightly weaker bound directly from Proposition 6.6(vi) of [6], which contains the expression

$$\exp\left(\frac{3\left|\log \frac{x}{4\pi} + \frac{i\pi}{2}\right| + 3.58}{x - 8.52}\right) \left(1 + \frac{1.24(3^y + 3^{-y})}{N - 0.125} + \frac{6.92}{x - 12}\right).$$

On the theorem domain $x \geq 200$ one has $x - 8.52 > x - 12 > 0$; applying $1 + u \leq e^u$ to the second factor and enlarging the denominator $x - 8.52$ to $x - 12$ yields the *conservative corollary*

$$e_{C,0} \leq \left(\frac{x}{4\pi}\right)^{-(1+y)/4} \exp\left(-\frac{t}{16} \log^2 \frac{x}{4\pi} + \frac{1.24(3^y + 3^{-y})}{N - 0.125} + \frac{3|\log \frac{x}{4\pi} + \frac{i\pi}{2}| + 10.50}{x - 12}\right), \quad (11)$$

with $3.58 + 6.92 = \frac{21}{2} = 10.50$ exactly. Every numerical error computation below (finite range, tail, and barrier) uses (11), and uses the denominator $x - 6.66$, which appears in equation (23) and in Proposition 6.6(iv),(v) of [6], for $e_A + e_B$.

Remark 2.2. Equation (24) of [6] displays the bound (11) with the constant 10.44 in place of 10.50. The sentence following Proposition 6.6 in [6] states that (24) is obtained from part (vi) of the proposition by “the inequality $1 + u \leq \exp(u)$ ” followed by the bound $\frac{1}{x-8.52} \leq \frac{1}{x-12}$; carrying out this recipe yields the constant $3.58 + 6.92 = 10.50$, and we are not aware of an argument in [6] producing the displayed 10.44. Accordingly, the current proof relies on (24): all error computations use only (11), which follows rigorously from Proposition 6.6(vi) by the two elementary steps just described, uniformly on the domain (7). (The weaker constant costs a factor $e^{0.06/(x-12)}$, which is utterly negligible at $x \approx 6 \times 10^{12}$.)

3 Hypothesis (i): the verified height

Platt and Trudgian [5, Theorem 1] proved that every nontrivial zero of ζ with ordinate in $(0, T_{\text{PT}}]$,

$$T_{\text{PT}} = 3000175332800,$$

lies on the critical line. (Their statement reads: “The Riemann hypothesis is true up to height 3000 175 332 800. That is, the lowest 12 363 153 437 138 non-trivial zeroes ρ have $\Re \rho = 1/2$.” The method — counting sign changes of the completed zeta function on the critical line and confirming by Turing’s method that all zeros are accounted for — yields the off-line exclusion form used here.) Hypothesis (i) of Theorem 2.1 requires the absence of zeros with $\sigma \geq \frac{1+y_0}{2} > \frac{1}{2}$ and $0 \leq T \leq X/2$. Since

$$\frac{X}{2} = 3000000092913.5, \quad T_{\text{PT}} - \frac{X}{2} = \frac{350479773}{2} > 0,$$

the verified region covers all ordinates $0 < T \leq X/2$ with margin $\approx 1.75 \times 10^8$. On the segment $T = 0$, $\frac{1}{2} < \sigma < 1$, one argues directly: the Dirichlet eta function satisfies $\eta(\sigma) = \sum_{n \geq 1} (-1)^{n-1} n^{-\sigma} > 0$ for $0 < \sigma < 1$ (group consecutive terms), while $\eta(\sigma) = (1 - 2^{1-\sigma})\zeta(\sigma)$ with $1 - 2^{1-\sigma} < 0$; hence $\zeta(\sigma) < 0 \neq 0$ there. At $\sigma = 1$, ζ has a pole. This supplies hypothesis (i).

As with any use of a computationally established theorem, this step inherits the correctness of the Platt–Trudgian computation itself (a rigorous interval-arithmetic computation, independent of anything in the present paper); we accept it here as a published-theorem input.

4 Hypothesis (ii), finite range: the window scan

Hypothesis (ii) concerns nonvanishing of H_{t_0} on

$$x \geq x_* := X + \sqrt{1 - y_0^2}, \quad y_0 \leq y \leq \sqrt{1 - 2t_0}. \quad (12)$$

Gomila's proof splits this region at the abscissa corresponding to the window index $N_* = 3840000$: the finite range $690988 \leq N \leq 3840000$ is handled in this section by 3,149,013 individually certified interval computations, and the infinite range $N \geq 3840000$ in Section 5 by a single analytic contraction argument. The two ranges overlap on a full window, eliminating endpoint risk.

4.1 Windows and the freeze lemma

At $t = t_0$ define

$$x_N = 4\pi\left(N^2 - \frac{t_0}{16}\right), \quad W_N = [x_N, x_{N+1}),$$

so that $x \in W_N$ iff the window index of x equals N . Exact rational Machin-type bounds for π and exact integer square-root brackets (independently confirmed by a 400-bit interval computation) give

$$x_{690988} < x_* < x_{690989}, \quad (13)$$

with margins exceeding 5.37×10^6 and 1.19×10^7 respectively; thus x_* lies strictly inside W_{690988} , and the half-open windows tile $[x_*, x_{3840001})$ with no gaps or overlaps.

The certified computations evaluate, for each N , the majorants of (10) *frozen at the left edge* $x = x_N$ of the window. The following elementary lemma justifies this.

Lemma 4.1 (Window freeze). *For $y_0 \leq y \leq \sqrt{1 - 2t_0}$ define*

$$G(x, y) = e^{y/50} \left(\frac{x}{4\pi}\right)^{-y/2}, \quad K(x, y) = \frac{t_0 y}{2(x-6)}, \quad \Sigma(x, y) = \frac{1+y}{2} + \frac{t_0}{4} \log \frac{x}{4\pi} - \frac{t_0}{2x^2} \left(1 - 3y + \frac{4y(1+y)}{x^2}\right)_+.$$

Then for every N in the finite range, every $x \in W_N$ and every y in the stated range,

$$G(x, y) \leq G(x_N, y), \quad K(x, y) \leq K(x_N, y), \quad \Sigma(x, y) \geq \Sigma(x_N, y).$$

Proof. $\partial_x G = -\frac{y}{2x} G < 0$ and $\partial_x K = -\frac{t_0 y}{2(x-6)^2} < 0$ give the first two inequalities. For the third, write $h(x, y) = 1 - 3y + A(y)/x^2$ with $A(y) = 4y(1+y) \geq 0$; then $h_x = -2A/x^3 \leq 0$, so h_+ is nonincreasing in x , and since x^{-2} is positive and decreasing, the product $x^{-2}h_+$ is nonincreasing. The logarithmic term is strictly increasing, so Σ is strictly increasing in x , including across the kink $h = 0$ (both one-sided derivatives there are positive). $\square$

Note that $e^{y/50} = e^{0.02y}$, matching (10); G, K, Σ frozen at x_N are precisely the quantities g_N (bound for $|\gamma|$), k_N (bound for $|\kappa|$) and σ_N (lower bound for $\operatorname{Re} s_*$) used in the definition of the certified quantities of Section 4.3.

4.2 The mollified lower bound for $|f_{t_0}|$

Fix a window N , a height $y > 0$, and a finite set of auxiliary primes $\mathcal{P}$ with $D = \prod_{p \in \mathcal{P}} p$. Let g_N, k_N, σ_N be as above, and introduce the real *Euler mollifier*

$$E(s_*) = \prod_{p \in \mathcal{P}} (1 - b_{t_0}(p) p^{-s_*}) = \sum_{d|D} \lambda_d d^{-s_*}, \quad \lambda_d = (-1)^{\omega(d)} \prod_{p|d} b_{t_0}(p) \in \mathbb{R}.$$

Define the convolved coefficients (for $1 \leq n \leq DN$)

$$B_{N,n} = \sum_{\substack{d|D, d|n \\ n/d \leq N}} \lambda_d b_{t_0}(n/d), \quad A_{N,n} = \sum_{\substack{d|D, d|n \\ n/d \leq N}} \lambda_d b_{t_0}(n/d) (n/d)^y,$$

so $B_{N,1} = A_{N,1} = 1$, and set

$$\begin{aligned} M_N &= \sum_{d|D} |\lambda_d| d^{-\sigma_N}, \quad C_N = \sum_{m=2}^N b_{t_0}(m) (m^{k_N} - 1) m^{y-\sigma_N}, \\ L_N &= \frac{1 - g_N - \sum_{n=2}^{DN} (|B_{N,n}| + g_N |A_{N,n}|) n^{-\sigma_N}}{M_N} - g_N C_N. \end{aligned} \quad (14)$$

Lemma 4.2 (Mollified lower bound). *If $L_N > 0$, then $|f_{t_0}(x + iy)| \geq L_N$ for all $x \in W_N$.*

Proof. Write (9) as $f_t = A + \gamma C_\kappa$ with $A = \sum_{m \leq N} b_t(m) m^{-s_*}$ and $C_\kappa = \sum_{m \leq N} m^y b_t(m) m^{-\bar{s}_* - \kappa}$. Exact Dirichlet convolution gives the two identities

$$E(s_*) A = \sum_{n=1}^{DN} B_{N,n} n^{-s_*}, \quad \overline{E(s_*)} C_0 = \sum_{n=1}^{DN} A_{N,n} n^{-\bar{s}_*}, \quad (15)$$

where the second identity uses that the λ_d are *real* (so that $\overline{E(s_*)} = \sum_d \lambda_d d^{-\bar{s}_*}$). By the triangle inequality and $\operatorname{Re} s_* \geq \sigma_N$,

$$|EA| \geq 1 - \sum_{n=2}^{DN} |B_{N,n}| n^{-\sigma_N}, \quad |\overline{E} C_0| \leq 1 + \sum_{n=2}^{DN} |A_{N,n}| n^{-\sigma_N}.$$

For complex κ with $|\kappa| \leq k_N$,

$$|m^{-\kappa} - 1| = |e^{-\kappa \log m} - 1| \leq e^{|\kappa| \log m} - 1 \leq m^{k_N} - 1, \quad (16)$$

whence $|\overline{E}(C_\kappa - C_0)| \leq |E| C_N$. Combining, and using $|\gamma| \leq g_N$,

$$|E| |f_t| \geq Q_N - g_N |E| C_N, \quad Q_N := 1 - g_N - \sum_{n=2}^{DN} (|B_{N,n}| + g_N |A_{N,n}|) n^{-\sigma_N}.$$

If $L_N > 0$ then $Q_N = M_N(L_N + g_N C_N) > 0$; the displayed inequality then forces $E \neq 0$ (otherwise $0 \geq Q_N > 0$). Dividing by $|E|$ and using $|E| \leq M_N$ together with $Q_N > 0$,

$$|f_t| \geq \frac{Q_N}{|E|} - g_N C_N \geq \frac{Q_N}{M_N} - g_N C_N = L_N. \quad \square$$

The point of Lemma 4.2 is that any certified per-window lower bound for L_N is *already* in the same normalized units as the additive error in (8); no further Euler-factor conversion intervenes.

4.3 The certified lower bounds

For each integer N with $690988 \leq N \leq 3840000$, define a set of auxiliary primes

$$\mathcal{P}(N) = \begin{cases} \{2, 3, 5, 7, 11\}, & 690988 \leq N \leq 728999, \\ \{2, 3, 5, 7\}, & 729000 \leq N \leq 818999, \\ \{2, 3, 5\}, & 819000 \leq N \leq 1027999, \\ \{2, 3\}, & 1028000 \leq N \leq 3840000, \end{cases} \quad (17)$$

and let L_N denote the quantity (14) formed with the prime set $\mathcal{P}(N)$, at $t = t_0$ and $y = y_0$, with the constants $g_N = G(x_N, y_0)$, $k_N = K(x_N, y_0)$, $\sigma_N = \Sigma(x_N, y_0)$ frozen at the window's left edge as in Lemma 4.1. (The number of terms in L_N grows with $|\mathcal{P}(N)|$; the choice (17) uses the larger, more effective mollifiers only on the ranges of N where they are needed.)

Proposition 4.3 (Certified lower bounds for L_N ; computer-assisted). *For every integer $690988 \leq N \leq 3840000$,*

$$L_N \geq T_{\min} = 791366 \times 10^{-12} > 0.$$

More precisely, the minimum of L_N over each of the four ranges in (17) is bounded below as listed in Table 1; the global bound $T_{\min}$ is attained at $N = 690988$.

$\mathcal{P}(N)$	range of N	certified lower bound for $\min L_N$
$\{2, 3, 5, 7, 11\}$	$690988 \leq N \leq 728999$	0.000000791366
$\{2, 3, 5, 7\}$	$729000 \leq N \leq 818999$	0.000315112481
$\{2, 3, 5\}$	$819000 \leq N \leq 1027999$	0.000305788816
$\{2, 3\}$	$1028000 \leq N \leq 3840000$	0.000309285478

Table 1: The auxiliary prime sets (17) and, for each of the four ranges of N , the certified lower bound for $\min_N L_N$ established by the verification run described in the proof of Proposition 4.3.

Proof. The proof is computer-assisted; the verification program `prop43_proof.c`, included with this manuscript as supplementary material (directory `code/prop43`), computes for each N an outward-rounded interval enclosure of a quantity provably at most L_N , and prints the floor of 10^{12} times the lower endpoint of that enclosure. Each printed value is therefore a certified lower bound for L_N . The program is derived from Gomila's source (as documented in its header) and was reviewed line by line as part of this audit. It has two evaluation modes: a direct term-by-term interval evaluation of (14), and a production sweep that majorizes the sum in (14) by activation-frozen weights (conservative because the factor g decreases in N) and evaluates the result through certified Taylor moments with an explicit Lagrange remainder. The review verified that every step of both modes is conservatively directed, and the two modes produce the identical bound at the extremal index $N = 690988$.

The program was compiled against an independently installed FLINT/Arb 3.6.0 and run at 256-bit precision over the four configurations of (17), in each case at $y = y_0$ and over a closed rational t -interval containing t_0 (for the first range, the singleton $\{129/800\}$). Every one of the 3,149,013 values of N produced a certified positive bound, with no uncertified rows; the concatenated outputs cover $690988 \leq N \leq 3840000$ with no gaps or duplicates;

and the certified per-range minima are as in Table 1, with global minimum 791366×10^{-12} at $N = 690988$. Per-range summaries (`runs/*_summary.txt`) and the complete per- N records (`runs/*_sweep.out`) are included in the supplementary material. $\square$

4.4 Transfer to the full height range

Proposition 4.3 concerns the single height $y = y_0$, while hypothesis (ii) requires nonvanishing for all $y \in [y_0, \sqrt{1-2t_0}]$. For fixed N write $L_N(y)$ for the quantity (14) with its explicit y -dependence restored (the constants g_N, k_N, σ_N frozen in x as before, but evaluated at height y ; likewise the coefficients $A_{N,n}$ and the sums M_N, C_N), so that $L_N = L_N(y_0)$. The bridge between Proposition 4.3 and the full height range is the following statement.

Proposition 4.4 (Height transfer; partly computer-assisted). *For every integer $690988 \leq N \leq 3840000$ and every $y \in [y_0, \sqrt{1-2t_0}]$,*

$$L_N(y) \geq L_N(y_0).$$

Gomila’s proof establishes three monotonicity statements, which together imply the proposition: as y increases, (a) the majorant sum

$$\mathcal{F}_{N,\mathcal{P}}(y) = g_N(y) + \sum_{n=2}^{DN} (|B_{N,n}| + g_N(y)|A_{N,n}(y)|) n^{-\sigma_N(y)}$$

subtracted in (14) is nonincreasing; (b) the mollifier bound $M_N(y)$ is positive and nonincreasing; and (c) the correction term $g_N(y)C_N(y) \geq 0$ is nonincreasing. Indeed, granting (a)–(c): the numerator $1 - \mathcal{F}_{N,\mathcal{P}}(y)$ is nondecreasing and is positive at y_0 (since $1 - \mathcal{F}_{N,\mathcal{P}}(y_0) = M_N L_N(y_0) + M_N g_N C_N \geq M_N T_{\min} > 0$ by Proposition 4.3), so the quotient $(1 - \mathcal{F}_{N,\mathcal{P}}(y))/M_N(y)$ is nondecreasing, and subtracting the nonincreasing nonnegative term $g_N(y)C_N(y)$ preserves this.

For statement (a), we bound the upper-right Dini derivative $D^+[(|B| + \gamma|A|)n^{-\sigma}]$ termwise, using

$$D^+\{\gamma|A|\} \leq \gamma(g|A| + |A'|), \quad g = \frac{\gamma'}{\gamma} = \frac{1}{50} - \frac{1}{2} \log\left(N^2 - \frac{t_0}{16}\right) < 0,$$

(valid also at zeros of A), together with the slope bound $\sigma_N(y_2) - \sigma_N(y_1) \geq \frac{1}{2}(y_2 - y_1)$. The termwise bound retains the negative term $\gamma g|A|$; discarding it (as an earlier, now-abandoned “standard majorant” shortcut did) is not sufficient. The signs of the finitely many divisor patterns are handled by an exhaustive case decomposition into $3^{|\mathcal{P}|}$ cells (243, 81, 27, 9 cells for the four prime sets in (17)), and rigorous Arb runs at 180 and 256 bits certify the required strict inequalities; the worst certified ratio is $\leq 0.99999860767275095 < 1$. Local Lipschitz continuity converts the pointwise upper-Dini bounds into global monotonicity. Statements (b) and (c) are certified by a separate interval computation: the worst certified logarithmic rate of decrease is $\leq -1.363 \cdots < 0$.

Gap / review point 1 (category A/B: the height transfer). This gap is the proof of Proposition 4.4 — the monotonicity statements (a)–(c), whose certified margins are small ($1 - \text{ratio} \approx 1.4 \times 10^{-6}$ at the worst cell of statement (a)). Review must confirm: the Dini-derivative calculus at zeros of $A_{N,n}$; correctness of the sign-pattern cell decomposition

and its exhaustiveness; the one-sided rounding of the cell endpoints; and the passage from pointwise Dini bounds to global monotonicity. (The deduction of Proposition 4.4 from (a)–(c) is the elementary argument given above.)

4.5 Error bounds and the finite-range conclusion

The remaining ingredient is a uniform bound on the errors of Theorem 1.3 over the finite region.

Proposition 4.5 (Uniform error bound, finite region; computer-assisted). *For all $x \geq x_*$ and $y_0 \leq y \leq \sqrt{1-2t_0}$, the error terms of Theorem 1.3 at $t = t_0$, bounded using equation (23) of [6] for $e_A + e_B$ and the conservative corollary (11) for $e_{C,0}$, satisfy*

$$e_A + e_B \leq 2.06 \times 10^{-12}, \quad e_{C,0} \leq 0.000000233492848188649183,$$

and hence

$$e_A + e_B + e_{C,0} \leq E_{\max} := 0.000000233494905212337849.$$

The proof is again an interval computation (the majorants are decreasing in x , so the enclosure is controlled by the left endpoint x_*); unlike Proposition 4.3, it has not yet been independently reproduced.

Gap / review point 2 (category A: the finite-region error computation). This gap is the computer-assisted proof of Proposition 4.5. In Gomila’s repository the computation is part of `verifiers/verify_finite_and_binding.py`; its verification requires reading that program’s error evaluation against equation (23) of [6] and the corollary (11), and re-running it — a small computation, in contrast with Proposition 4.3, since the majorants are monotone in x and only the extremal enclosures enter.

We can now state the conclusion of the finite range.

Proposition 4.6 (Nonvanishing at time t_0 , finite range). *$H_{t_0}(x + iy) \neq 0$ for all $x_* \leq x < x_{3840001}$ and $y_0 \leq y \leq \sqrt{1-2t_0}$.*

Proof. Let x be in the stated range and let N be its window index, so that $690988 \leq N \leq 3840000$ by (13), and $x \in W_N$. For y in the stated range, Lemma 4.2 (applied with the height y , the prime set $\mathcal{P}(N)$, and the frozen constants of Lemma 4.1), together with Propositions 4.3 and 4.4, gives

$$|f_{t_0}(x + iy)| \geq L_N(y) \geq L_N(y_0) \geq T_{\min}.$$

Combining with Proposition 4.5,

$$|f_{t_0}(x + iy)| - (e_A + e_B + e_{C,0}) \geq T_{\min} - E_{\max} \geq 0.000000557871094787 > 0. \quad (18)$$

The point (x, y, t_0) lies in the region (7), so [6, Corollary 1.4] (the criterion for non-vanishing quoted in Section 2.2) yields $H_{t_0}(x + iy) \neq 0$. $\square$

5 Hypothesis (ii), infinite tail

This section proves nonvanishing of H_t for *all* windows $N \geq N_* = 3840000$, all $t \in I_t = [\frac{129}{800}, \frac{161250001}{10^9}]$ (a closed interval whose left endpoint is exactly t_0), and all $y \in [y_0, \sqrt{1-2t}]$, by a single interval computation at the cutoff N_* combined with monotonicity arguments in N and y . Since $t_0 \in I_t$, this in particular covers the case $t = t_0$ needed for hypothesis (ii); the last window $W_{3840000}$ of the finite range overlaps the tail's starting abscissa $x_{3840000}$, so the two ranges jointly cover all $x \geq x_*$.

5.1 Setup: mollifier and frozen exponents

Fix the sparse mollifier support $\mathcal{S} = \{1, 2, 3, 4, 5, 6, 7, 10, 11, 13, 14\}$ and the exact rational vector

d	1	2	3	4	5	6	7	10	11	13	14
λ_d	1	$-\frac{1021}{1000}$	$-\frac{1054}{1000}$	$-\frac{9}{200}$	$-\frac{1119}{1000}$	$\frac{1001}{1000}$	$-\frac{1043}{1000}$	$\frac{128}{125}$	$-\frac{161}{200}$	$-\frac{447}{500}$	$\frac{456373}{500000}$

($\lambda_1 = 1$; $\sum_d |\lambda_d| = 9.918746$ exactly), and set $M_\lambda(z) = \sum_{d \in \mathcal{S}} \lambda_d d^{-z}$. (Unlike the Euler-product mollifier of Section 4.2, this vector was chosen by numerical optimization; its only structural requirement is $\lambda_1 = 1$ and real entries.) Fix also the head lengths $M = 153814$ and $M_{\text{err}} = 3000$.

For $x \geq X_N(t) := 4\pi(N^2 - \frac{t}{16})$ with window index $N \geq N_*$, define frozen exponents (uniform over I_t , via rigorous upper bounds $\hat{\delta} \geq \delta(N_*, t)$, $\hat{k} \geq k(N_*, t)$ for the quantities $\delta(N, t) = \frac{t}{4}[-\log(1 - \frac{t}{16N^2})] + \frac{t}{2X_N(t)^2}$ and $k(N, t) = \frac{t}{2(X_N(t)-6)}$, both decreasing in N):

$$\sigma_1 = \frac{1+y}{2} + \frac{t}{2} \log N - \hat{\delta}, \quad \sigma_2 = \frac{1-y}{2} + \frac{t}{2} \log N - \hat{\delta} - \hat{k}, \quad G = e^{0.02y} \left(N^2 - \frac{t}{16}\right)^{-y/2}.$$

By (10) these satisfy $|\gamma| \leq G$, $\text{Re } s_* \geq \sigma_1$, $|\kappa| \leq \hat{k}$ throughout the window, and σ_2 lower-bounds the effective exponent of the second (“A-side”) sum in (9) after the factors n^y and $n^{-\kappa}$ are absorbed.

Gap / review point 3 (category B: the frozen-exponent reductions). The derivation of σ_1, σ_2 from (10) — in particular the claim that the positive-part term in (10) is at most 1 on the tail domain, and the identity $\frac{t}{4} \log(N^2 - \frac{t}{16}) = \frac{t}{2} \log N - \frac{t}{4}[-\log(1 - \frac{t}{16N^2})]$ — is elementary but must be checked, as must the claim that all three window-edge evaluations are conservative (“worst at the left edge”).

5.2 The endpoint-cap lemma

The analytic engine for tail sums is:

Lemma 5.1 (Endpoint cap). *For $1 \leq a < c$ set $E_{t,\sigma}(u) = \exp((1-\sigma) \log u + \frac{t}{4} \log^2 u)$ and*

$$\text{Cap}_t(a, c; \sigma) = \max(E_{t,\sigma}(a), E_{t,\sigma}(c)) \log \frac{c}{a}.$$

If $\sigma > \frac{t}{2} \log c$, then for all integers $a \leq J < K \leq c$,

$$\sum_{J < n \leq K} b_t(n) n^{-\sigma} \leq \text{Cap}_t(a, c; \sigma).$$

Proof. The logarithmic derivative of $b_t(u)u^{-\sigma}$ in $\log u$ is $\frac{t}{2}\log u - \sigma < 0$ on $[a, c]$, so the summand decreases and the sum is at most $\int_a^c b_t(u)u^{-\sigma}du$. Substituting $v = \log u$ makes the integrand $\exp((1 - \sigma)v + \frac{t}{4}v^2)$, which is log-convex in v , hence maximized at an endpoint of $[\log a, \log c]$; multiplying by the length $\log(c/a)$ of the v -interval gives the bound. $\square$

5.3 Exact convolution and the contraction D

Define, for $N \geq N_*$ and (x, y, t) with x in the window of index N ,

$$P = \sum_{m=2}^M |c_m(t)| m^{-\sigma_1}, \quad c_m(t) = \sum_{\substack{d \in \mathcal{S} \\ d|m}} \lambda_d b_t(m/d), \quad M_{\max} = \sum_{d \in \mathcal{S}} |\lambda_d| d^{-\sigma_1},$$

$$\text{TR} = \sum_{d \in \mathcal{S}} |\lambda_d| d^{-\sigma_1} \text{Cap}_t(\lfloor M/d \rfloor, N; \sigma_1), \quad \text{AB} = G\left(\sum_{k \leq M} b_t(k) k^{-\sigma_2} + \text{Cap}_t(M, N; \sigma_2)\right),$$

together with a further nonnegative term $\text{OV} \geq 0$ (an “overshoot” cap that Gomila’s implementation adds; see the remark after the proof), and set

$$D = D(N, t, y) := P + \text{TR} + \text{OV} + M_{\max} \text{AB}. \quad (19)$$

Lemma 5.2 (Mollified contraction bound). *For every $N \geq N_*$, $t \in I_t$, $0 < y \leq 1$ and x in the window of index N ,*

$$|M_\lambda(s_*)f_t(x + iy) - 1| \leq D.$$

Consequently, if $D < 1$ then

$$|f_t(x + iy)| \geq \frac{1 - D}{M_{\max}}. \quad (20)$$

Proof. Write the two sums in (9) as F_B and F_A ($f_t = F_B + \gamma F_A$). Since $M < N_* \leq N$, multiplying F_B by the mollifier and partitioning the index pairs (d, k) by $dk \leq M$ versus $dk > M$ gives the *exact* identity

$$M_\lambda(s_*)F_B - 1 = \sum_{m=2}^M c_m(t) m^{-s_*} + \sum_{d \in \mathcal{S}} \lambda_d d^{-s_*} \sum_{\lfloor M/d \rfloor < k \leq N} b_t(k) k^{-s_*}, \quad (21)$$

using $c_1 = \lambda_1 b_t(1) = 1$. By $\text{Re } s_* \geq \sigma_1$, the head is bounded by P and, by Lemma 5.1, each inner tail sum by $\text{Cap}_t(\lfloor M/d \rfloor, N; \sigma_1)$, so the second term is bounded by TR . On the A -side, the frozen-exponent bounds give $|\gamma F_A| \leq \text{AB}$ (finite head plus Lemma 5.1 for the tail, at exponent σ_2), while $|M_\lambda(s_*)| \leq M_{\max}$; hence $|M_\lambda(s_*)\gamma F_A| \leq M_{\max} \text{AB}$. Adding the nonnegative OV only weakens the bound, and the first claim follows. For the second: $|M_\lambda(s_*)f_t| \geq 1 - D > 0$ and $|M_\lambda(s_*)| \leq M_{\max}$, so $|f_t| \geq (1 - D)/M_{\max}$. $\square$

Remark 5.3. The term OV plays no role in the mathematics — the identity (21) is fully covered by $P + \text{TR}$ — but Gomila’s program includes it in D , so it is included in (19) in order that the certified numerical bound on D (Proposition 5.5 below) majorizes the same quantity the program computes.

Gap / review point 4 (category B: the convolution partition and OV). Identity (21) requires the pair partition ($dk \leq M$ vs. $dk > M$, with $dk > M \iff k > \lfloor M/d \rfloor$) to be disjoint and exhaustive, and requires $M < N$ so that all head indices $m \leq M$ arise only from $k \leq N$. The proof asserts that the extra term OV is pure nonnegative padding, not a double-counted or missing piece of (21); because a mistaken accounting here would silently break (19), a referee should re-derive (21) and confirm that TR alone covers the entire remainder.

5.4 Reduction to the cutoff

The interval computation of Section 5.6 evaluates D , and the error majorants, only at the cutoff $N = N_*$, over the closed intervals

$$I_{\text{box}} = \left[\frac{1872719}{10^7}, \frac{23409}{125000} \right], \quad I_{\text{ext}} = \left[\frac{1872719}{10^7}, \frac{8231039}{10^7} \right],$$

where I_{box} contains y_0 and the top of I_{ext} is at least $\sqrt{1-2t}$ for every $t \in I_t$. The reduction from this single evaluation to all $N \geq N_*$ and the full height range is the content of the next lemma. In its statement, the *cutoff conditions* are the following inequalities, all evaluated at $N = N_*$, uniformly for $t \in I_t$:

$$\frac{t}{2} \log a (\log N_* - \log a) > 1 \quad \text{and} \quad (\sigma - 1)(\log N_* - \log a) > 1 \quad (22)$$

for each cap $\text{Cap}_t(a, \cdot; \sigma)$ appearing in D and in the error bounds (the left endpoints $a = \lfloor M/d \rfloor$ for $d \in \mathcal{S}$, $a = M$, and $a = M_{\text{err}}$, with $\sigma = \sigma_1$ or σ_2 as appropriate), together with the A -side slope condition

$$0.02 - \frac{1}{2} \log(N_*^2 - \frac{t}{16}) + \frac{1}{2} \log N_* < 0. \quad (23)$$

Lemma 5.4 (Reduction to the cutoff). *Assume the cutoff conditions (22)–(23) hold. Then for every $N \geq N_*$, every $t \in I_t$ and every $y \in [y_0, \sqrt{1-2t}]$,*

$$D(N, t, y) \leq \sup_{y' \in I_{\text{box}}} D(N_*, t, y'),$$

and the error majorants of Section 5.5 at any $x \geq X_{N_}(t)$ and $y \in [y_0, \sqrt{1-2t}]$ are bounded by their suprema over $\{N_*\} \times I_t \times I_{\text{ext}}$ evaluated at the window's left edge.*

Proof (outline, from the candidate materials). Write $q = \log N$; the frozen exponents satisfy $d\sigma_i/dq = t/2 > 0$. For a cap with fixed left endpoint a , each of the two candidate endpoint values of $\text{Cap}_t(a, N; \sigma)$, multiplied by the length $q - \log a$, has negative logarithmic q -derivative provided (22) holds at N ; and both conditions in (22), once true at N_* , persist for all larger N (in the first, both factors increase with q ; in the second, σ increases with q). The head P , the mollifier bound M_{max} , the prefactor G , the finite A -head, and the moving cap inside OV (whose endpoints both scale with N) all decrease in N as well, each factor being checked in its conservative direction; sums and products of nonnegative decreasing quantities decrease, so $D(N, t, y) \leq D(N_*, t, y)$.

In the height variable, $\partial_y \sigma_1 = \frac{1}{2}$ makes all B -side terms nonincreasing in y ; on the A -side the combined factor $G E_{t, \sigma_2}(u)$ has logarithmic y -derivative at most the left side of (23), which

is negative and only decreases as N grows. Hence $D(N_*, t, \cdot)$ is nonincreasing above I_{box} , and since $y_0 \in I_{\text{box}}$, every $y \in [y_0, \sqrt{1-2t}]$ is either in I_{box} or above it. The error majorants are handled in the same way (Section 5.5). No sampling in N occurs, and no monotonicity in t is used: the whole interval I_t is enclosed in the computation. $\square$

Gap / review point 5 (category B: the reduction to the cutoff). The proof of Lemma 5.4 — the persistence of (22) in N ; the treatment of the moving cap in OV; the y -derivative computation for the A -side; and the claim that the individually-monotone pieces compose to monotonicity of D itself — constitutes the mathematical heart of the tail argument and must be verified line by line. The interplay between quantities frozen at N_* (uniform in I_t) and genuinely N -dependent quantities deserves particular care.

5.5 Tail error terms

The tail error computation bounds $e_A + e_B$ from the exact definitions (71)–(72) of [6] (sharper than display (23), which applies the worst-case factor $N^{|\kappa|}$ to every term: (71) carries the per-term factor $n^{-\Re \kappa}$, bounded termwise by $n^{\hat{\kappa}}$) via the quantity $\Delta(N, t) = ((t^2/16) \log^2(N^2 - \frac{t}{16}) + 0.626)/(X_N(t) - 6.66)$, and bounds $e_{C,0}$ by the conservative corollary (11) with $\log(N^2 - \frac{t}{16})$ substituted for $\log \frac{x}{4\pi}$. Elementary derivative computations (reproduced in the candidate’s tail note) show both bounds are maximal at each window’s left edge and decreasing in N ; this is the error half of Lemma 5.4.

5.6 The certified cutoff inequalities and the tail theorem

The computational input of the tail argument is the following.

Proposition 5.5 (Certified cutoff inequalities; computer-assisted). *The cutoff conditions (22)–(23) hold, and*

$$\sup_{t \in I_t, y \in I_{\text{box}}} D(N_*, t, y) < 0.999721, \quad M_{\max} < 1.608290,$$

$$\sup (e_A + e_B + e_{C,0}) < 1.1672 \times 10^{-8},$$

the last supremum being over $\{N_*\} \times I_t \times I_{\text{ext}}$ in the sense of Lemma 5.4. In particular

$$\frac{1-D}{M_{\max}} - (e_A + e_B + e_{C,0}) > 0.00017352 > 0.$$

The proof is a standalone FLINT/Arb computation run at both 256 and 512 bits of precision, with a separate Python interval implementation as cross-corroboration.

Gap / review point 6 (category A: the tail computation). This gap is the computer-assisted proof of Proposition 5.5: the stated bounds are machine output of a single C program (`verify_tail_arb.c`). The margin $1 - D < 0.000279$ is small: the contraction only barely closes. The program’s proof-to-code map (Section 10 of the Gomila’s tail note) should be audited check-by-check, and the computation reproduced with an independently built FLINT/Arb.

Theorem 5.6 (Tail theorem). *For every $t \in I_t$, $y \in [y_0, \sqrt{1-2t}]$ and $x \geq X_{N_*}(t)$,*

$$|f_t(x+iy)| > e_A + e_B + e_{C,0},$$

and consequently $H_t(x+iy) \neq 0$ on this region.

Proof. Let N be the window index of x , so $N \geq N_*$. By Lemma 5.2, Lemma 5.4 and Proposition 5.5,

$$|f_t(x+iy)| \geq \frac{1 - D(N, t, y)}{M_{\max}} \geq \frac{1 - 0.999721}{M_{\max}} > 0.0001735 > 1.1672 \times 10^{-8} \geq e_A + e_B + e_{C,0},$$

using in the last step the error half of Lemma 5.4 together with Proposition 5.5. The point (x, y, t) lies in the region (7), so [6, Corollary 1.4] gives $H_t(x+iy) \neq 0$. $\square$

6 Hypothesis (iii): the barrier certificate

6.1 Statement, and reduction to a closed rectangle

Throughout this section let

$$R = [X, X+1] + i \left[\frac{1809}{10000}, 1 \right].$$

The goal of the section is the following theorem, whose proof is assembled in Section 6.5.

Theorem 6.1 (Barrier theorem). *$H_t(z) \neq 0$ for all $z \in R$ and $0 \leq t \leq t_0$.*

Theorem 6.1 implies hypothesis (iii) of Theorem 2.1: the curved barrier region of that hypothesis satisfies, horizontally, $X \leq x \leq X + \sqrt{1-y_0^2} \leq X+1$; and vertically, $y \geq \sqrt{y_0^2 + 2(t_0 - t)} \geq y_0 > \frac{1809}{10000}$ (indeed $y_0^2 - \left(\frac{1809}{10000}\right)^2 = \frac{234599}{10^8} > 0$) and $y \leq \sqrt{1-2t} \leq 1$; hence it is contained in $R \times [0, t_0]$.

6.2 Uniform effective error on the box, including $t = 0$

Proposition 6.2 (Uniform error bound on the barrier box; computer-assisted). *For all $x \in [X, X+1]$ and $0 \leq t \leq t_0$, the window index satisfies $N = \lfloor \sqrt{\frac{x}{4\pi} + \frac{t}{16}} \rfloor = 690988$ and $N^2 < x/(4\pi)$. Moreover, for all $z \in R$ and $0 < t \leq t_0$, the error terms of Theorem 1.3, bounded with the constants of Section 2.3, satisfy*

$$e_A + e_B + e_{C,0} < 0.000356523011600040 < 0.00125. \quad (24)$$

Gomila's proof is a 256-bit Arb enclosure over the full closed box; the constancy of N makes f_t a single analytic expression on R , and $N^2 < x/(4\pi)$ fixes the sign of the heat exponent and yields the uniform bound $|\gamma|n^y \leq e^{0.02}$ for $1 \leq n \leq N$. The round number $0.00125 = \frac{1}{800}$ is the *approximation allowance* used by the prism criterion below; (24) says it is safely conservative. Verification of this computation is part of gap 10.

Theorem 1.3 is stated in [6] for $t > 0$; the following lemma extends its conclusion to the closed endpoint $t = 0$ on the compact box.

Lemma 6.3 (Extension to $t = 0$). *For all $z \in R$ and all $0 \leq t \leq t_0$ (including $t = 0$), $B_t(z) \neq 0$ and*

$$|g_t(z) - f_t(z)| \leq 0.00125,$$

where $g_t = H_t/B_t$ and f_t is the sum (9) with the constant window index $N = 690988$.

Proof. For $0 < t \leq t_0$ this is Theorem 1.3 together with Proposition 6.2. For the endpoint, fix $z \in R$ and argue by a limit $t_j \downarrow 0$: (1) by dominated convergence in (1) (majorant $e^{t_0 u^2 + u} |\Phi(u)|$, integrable by the super-exponential decay of Φ), $H_t(z)$ is jointly continuous down to $t = 0$ on R ; (2) $B_t^{\pm 1}$ is continuous and $B_t \neq 0$ on $R \times [0, t_0]$, because the argument $\frac{1+y-ix}{2}$ of M_0 stays away from the branch cut (Section 2.2) and the deforming exponential never vanishes; (3) since the window index is constant, f_t is a fixed finite sum of functions continuous in t ; (4) the error majorant is continuous in t and bounded by 0.00125 on $(0, t_0]$ by (24). Passing to the limit in $|g_{t_j} - f_{t_j}| \leq 0.00125$ gives the claim at $t = 0$. $\square$

Gap / review point 7 (category B: the $t = 0$ extension). The dominated-convergence proof of Lemma 6.3 is plausible but touches the statement boundary of the cited theorem. Review should confirm each of the four continuity claims (in particular claim (2)), and that the limiting inequality is inherited in the closed form claimed.

6.3 Uniform derivative bounds on the box

The moving-mesh argument below needs uniform derivative bounds on R , supplied by the following lemma.

Lemma 6.4 (Uniform derivative bounds). *For each $0 \leq t_i \leq t_{i+1} \leq t_0$ there are explicit, interval-computable quantities $D_z = D_z(t_i)$ and $D_t = D_t(t_i, t_{i+1})$ — defined by evaluating each factor of the pointwise derivative majorants of Lemma 8.4 of [6] at a conservative corner of the box, as described below — such that*

$$D_z \geq \sup_{z \in R} |\partial_z f_{t_i}(z)|, \quad D_t \geq \sup_{z \in R, t \in [t_i, t_{i+1}]} |\partial_t f_t(z)|.$$

Lemma 8.4 of [6] supplies *pointwise* majorants for these derivatives. Gomila’s proof of Lemma 6.4 upgrades them to uniform bounds on the closed box by endpoint separation: each factor of each majorant is classified as increasing or decreasing in x and in y , and evaluated at its own conservative corner (e.g. $\log(x/4\pi)$ at $X + 1$; reciprocals $1/(x - 6)$ at X ; the “A-core” $e^{0.02y} q(x)^{-y/2} N^{ty/(2(x-6))} n^y n^{-\text{Re } s^*}$ at $y = y_{\min}$, justified by an Arb-certified strict negativity of its logarithmic y -derivative). The proof also repairs two numerical-analysis interfaces of the original Polymath barrier code:

- the discrete sums over $1 \leq n \leq N$ in Lemma 8.4 are bounded by an exact 16-term head plus $\int_{16}^N F(u) du$, with fail-closed checks certifying that the continuous summand decreases for $u \geq 16$ (the naive bound $F(1) + \int_1^N F$ used previously had no justification, the summand not being monotone near $n = 1$);
- the quadrature callbacks retain the full complex integration variable and use analytic logarithm/power operations when FLINT’s integrator requests holomorphic enclosures

(projecting to the real part, as an earlier implementation did, voids the Gauss–Legendre error bound).

Lemma 8.4 is stated for $t > 0$; since for fixed N the sums and their displayed derivatives are C^1 on the compact box, the uniform majorants (independent of ε) extend from $t \in [\varepsilon, t_0]$ to $t = 0$ by continuity.

Gap / review point 8 (category B: the derivative bounds). This gap is the proof of Lemma 6.4. The factor-by-factor monotonicity classification (including the positive-part kink in $\operatorname{Re} s_*$, the coupled decrease of the A -core in x , and the bracket-positivity conditions such as $2 \log N < \log(X/4\pi)$ and $c_0 > \frac{1}{2} \log N$) is intricate and entirely load-bearing for the barrier: an error in a single direction would invalidate D_z or D_t . The lemma, and its transcription into `TloopSinglemat_closed_cert.c`, require detailed review.

6.4 The prism criterion

The time interval $[0, t_0]$ will be covered by finitely many closed “prisms” $R \times [t_i, t_{i+1}]$. The following lemma reduces nonvanishing of H on a prism to finitely many verifiable conditions at its initial time t_i .

Lemma 6.5 (Prism criterion). *Let $0 \leq t_i < t_{i+1} \leq t_0$, and let $z_1, \dots, z_m$ (cyclically ordered, $z_{m+1} = z_1$) be a mesh on ∂R with consecutive spacing at most h . Let $D_z = D_z(t_i)$ and $D_t = D_t(t_i, t_{i+1})$ be as in Lemma 6.4, and suppose:*

(a) $|f_{t_i}(z_j)| \geq M_i$ for all j , where

$$M_i > \frac{D_z h}{2} + D_t (t_{i+1} - t_i) + 0.00125; \quad (25)$$

(b) *each argument increment $\arg(f_{t_i}(z_{j+1})/f_{t_i}(z_j))$ lies strictly in $(-\pi, \pi)$, and the total winding of the polygon $(f_{t_i}(z_j))_j$ about 0 is 0.*

Then $H_\tau(z) \neq 0$ for all $z \in R$ and $\tau \in [t_i, t_{i+1}]$.

Proof. On each half of a mesh subedge, both the true image curve $F(s) = f_{t_i}(\text{subedge}(s))$ and its endpoint chord lie in the closed disk of radius $D_z h/2$ about the nearer endpoint value (this is the factor $\frac{1}{2}$ in (25)); each disk is convex, so the straight-line homotopy from the true boundary curve to the mesh polygon stays inside these disks. Moving to time $\tau \in [t_i, t_{i+1}]$ displaces values by at most $D_t(\tau - t_i)$, and passing from f_τ to $g_\tau = H_\tau/B_\tau$ by at most 0.00125 (Lemma 6.3). Condition (25) says every resulting disk, centered at a mesh value of modulus $\geq M_i$, excludes the origin; hence the entire three-stage homotopy (polygon $\rightarrow$ true f_{t_i} -boundary image $\rightarrow f_\tau(\partial R) \rightarrow g_\tau(\partial R)$) avoids zero. By (b) the polygon has winding number 0 about the origin, so the zero-avoiding homotopy gives winding number 0 for $g_\tau(\partial R)$ as well, for every $\tau \in [t_i, t_{i+1}]$. Since $B_\tau \neq 0$ (Lemma 6.3), g_τ and H_τ have the same zeros on R , and g_τ is zero-free on ∂R ; the argument principle then gives $H_\tau \neq 0$ on R . $\square$

Gap / review point 9 (category B: the homotopy/winding argument). This gap is the proof of Lemma 6.5, which compresses several steps that deserve expansion in review: (a) that the polygon winding is well-defined and zero given the certified argument increments; (b) that a zero-avoiding homotopy between closed curves preserves winding number (standard, but the “nearer endpoint disk” bookkeeping on half-subedges, with time motion and approximation error added to the radius, should be checked against (25) carefully — e.g., whether M_i being a *minimum over mesh points* suffices when the relevant disk is centered at the nearer mesh value); and (c) that the argument principle applies on the closed rectangle (zero-freeness of H_τ on ∂R is part of the conclusion, not an assumption).

6.5 The certified prism decomposition

The computational input of the barrier argument is the following.

Proposition 6.6 (Certified prism decomposition; computer-assisted). *There exist times $0 = \tau_0 < \tau_1 < \dots < \tau_{883}$ with $\tau_{883} \geq t_0$ such that for each $0 \leq i < 883$, the hypotheses (a)–(b) of Lemma 6.5 hold for the prism $R \times [\tau_i, \min(\tau_{i+1}, t_0)]$, with certified margin in (25) at least 0.5198498946 in every prism.*

Gomila’s proof is the sealed 883-prism Arb computation. Its recorded data comprise: consecutive prisms with byte-identical adjacent time endpoints, first time exactly 0, final endpoint enclosing $\frac{129}{800}$; at each initial time, mesh enclosures of f_{τ_i} on ∂R , the derivative bounds of Lemma 6.4 with D_t interval-evaluated on the entire proposed prism (not sampled at its initial time), and the polygon conditions of Lemma 6.5(b), the aggregate winding enclosure being $[-8.95, 8.95] \times 10^{-13}$; any overlapping or indeterminate interval comparison counts as failure. Two full transcripts from distinct toolchains (Linux/GCC/FLINT 3.0.1 and macOS/Clang/FLINT 3.6.0) pass a 54-check fail-closed parser. The 62×62 complex coefficient matrix used to evaluate f_t is stored to 20 decimal digits; the verifier restores each component as a ball of radius $10^{-20} \max(1, |\text{component}|)$, independently regenerates all 7688 components as Arb balls (all contained, worst radius-to-allowance ratio < 0.018), and bounds the omitted Taylor tail by $1.955 \times 10^{-22} < 10^{-20}$, propagating the 10^{-20} radius through every value. (Restricting the final prism to its intersection with $[0, t_0]$ only shrinks the displacement terms in (25), so the certificate for the full final prism covers the truncated one.)

Gap / review point 10 (category A: the barrier computation). This gap is the computer-assisted proof of Proposition 6.6 together with Proposition 6.2: the prism certificate, the coefficient regeneration, and the uniform-error enclosure are machine output whose correctness rests on the C sources and FLINT/Arb. The repository provides a containerized fresh-replay path (`scripts/run_barrier_replay.sh`) that an independent referee should execute on an independently built toolchain. The provenance of the 62×62 coefficient matrix (that it is the Taylor-coefficient matrix of the intended function f_t , not merely a self-consistent regeneration) also requires review.

Proof of Theorem 6.1. By Proposition 6.6 and Lemma 6.5, $H_\tau(z) \neq 0$ for $z \in R$ and $\tau \in [\tau_i, \min(\tau_{i+1}, t_0)]$, for each i . The union of these intervals is $[0, t_0]$. $\square$

7 Assembly: proof of Theorem 1.1

Proof of Theorem 1.1 (candidate argument). The parameters (4) satisfy $0 < t_0 < \frac{1}{2}$ and $0 < y_0^2 < 1 - 2t_0$. Hypothesis (i) of Theorem 2.1 holds by Section 3, since $X/2 < T_{PT}$. Hypothesis (ii) holds by combining Proposition 4.6 (nonvanishing of H_{t_0} for $x_* \leq x < x_{3840001}$, $y_0 \leq y \leq \sqrt{1 - 2t_0}$) with Theorem 5.6 applied at $t = t_0 \in I_t$ (nonvanishing for $x \geq x_{3840000} = X_{N_*}(t_0)$): the two ranges overlap on the full window $W_{3840000}$, so together they cover all $x \geq x_*$. Hypothesis (iii) holds by Theorem 6.1, since the curved barrier region is contained in $R \times [0, t_0]$ (Section 6). Theorem 2.1 then gives

$$\Lambda \leq t_0 + \frac{y_0^2}{2} = \frac{893927}{5000000} = 0.1787854. \quad \square$$

8 Summary of gaps and items requiring verification

For the reader's convenience we collect the numbered gap / review points marked throughout the paper, grouped by category. *None* of these is presented by the candidate as a known error; each is a step whose validation is the task of the ongoing review, and an adverse finding at any *load-bearing* point could invalidate Theorem 1.1. (The published theorem inputs themselves are not listed here: as explained in Section 1.3, the statements cited from [6] and [5] have been checked against the arXiv sources and are applied on domains where their hypotheses hold, the one caveat being Remark 2.2. Items resolved during the review are also removed from this list; in particular, the computer-assisted proof of Proposition 4.3 has been reproduced on an independent toolchain and its program reviewed — see the proof given in Section 4.3.)

Category B: analytic arguments

1. (also A) The proof of Proposition 4.4 — the monotonicity statements (a)–(c) of Section 4.4; the worst certified ratio is within 1.4×10^{-6} of failing.
3. The frozen-exponent reductions opening the tail argument (Section 5).
4. The proof of Lemma 5.2: the convolution partition (21) and the role of the OV term.
5. The proof of Lemma 5.4: the cutoff conditions (22)–(23) and their persistence in N , the moving caps, and the composition of monotonicities.
7. The proof of Lemma 6.3 (the dominated-convergence extension of Theorem 1.3 to $t = 0$), and the analogous extension of Lemma 8.4 of [6].
8. The proof of Lemma 6.4: every monotone-direction classification, the head-plus-tail sum majorization, and the holomorphic quadrature contract.
9. The proof of Lemma 6.5: the half-subedge disk bookkeeping behind the factor $\frac{1}{2}$ in (25), the winding-number arguments, and the application of the argument principle on the closed rectangle.

Category A: machine computations

2. The computer-assisted proof of Proposition 4.5 (the uniform error bound $E_{\max}$ over the finite region; a small computation, not yet independently reproduced).
6. The computer-assisted proof of Proposition 5.5 ($1 - D < 2.8 \times 10^{-4}$; post-error margin $\approx 1.7 \times 10^{-4}$).
10. The computer-assisted proofs of Propositions 6.2 and 6.6 (the 883-prism certificate, coefficient regeneration, and uniform-error enclosure; comfortable margins, but correctness rests on the C sources and FLINT/Arb).

All category-A items should be reproduced on independently built toolchains via the repository's replay scripts; agreement of replays is evidence of reproducibility but cannot by itself validate the analytic reductions (category B) that give the computed quantities their meaning.

Overall assessment of proof architecture

The theorem inputs are settled: the criterion, the effective approximation, and the verified-height input are exactly as the candidate uses them, and the open questions are confined to the candidate's own lemmas (category B) and the machine computations (category A). Modulo those remaining points, the logical architecture is clean: three independent suppliers feed the three hypotheses of a single published criterion, the finite and tail ranges overlap on a full window, the barrier rectangle strictly contains the curved barrier, and every numerical acceptance condition is a strict directed interval inequality with fail-closed parsing. The thinnest quantitative margins are the finite-range post-error margin (5.6×10^{-7}), the height-transfer ratio ($1 - \text{ratio} \approx 1.4 \times 10^{-6}$), and the tail contraction ($1 - D \approx 2.8 \times 10^{-4}$); these are the natural places where a sign or normalization error in a category-B argument could tip a true margin negative, and they should receive the closest scrutiny.

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